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Galois Falsification: The Falsifiability Boundary of SCX

Galois Falsification — The Falsifiability Boundary of SCX
and the A_5 Experimental Protocol

SCX

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Abstract

Galois-SCX (SCX, 2026) proves a striking implication—**unsolvable audit group** $\Rightarrow$ **unauditable claims exist**. This paper extracts the falsifiable core: (1) we identify a single, concrete, experimentally testable prediction whose confirmation would refute SCX; (2) we design an experimental protocol to test this prediction, making SCX genuinely falsifiable—the **modus tollens** of the theory; (3) we provide two additional falsifiable predictions: (a) the A_5 performance obstruction, and (b) the normalizer violation test; (4) we construct an explicit $n = 5$ expert disagreement configuration that realizes the A_5 obstruction. Finally, an Honest Strike addresses the central question: **can A_5 genuinely be realized in practice?**

Keywords: Falsifiability; Galois correspondence; audit unsolvability; A_5 ; Popperian test; SCX; modus tollens

Contents

1 Introduction: Is SCX Falsifiable?

1.1 The Popperian Challenge

Popper [?] defined science by falsifiability: a theory must make predictions that could in principle be shown false. SCX’s core results—the Honest Person Theorem (Thm 3), the Strong Man Theorem (Thm 2), Situs Lipschitz [?, ?]—culminate in Galois-SCX [?].

But is SCX falsifiable?

SCX is a mathematical theory, and mathematical theorems, once proved, are not “falsifiable” in the empirical sense. However, Galois-SCX makes a **contingent claim** about the real world: that certain expert configurations realize unsolvable audit groups. This claim *is* falsifiable.

1.2 The Logical Structure

The core of Galois-SCX (Theorem 4) states:

T 1 (Unsolvability — Galois-SCX). *Let $\mathfrak{G}_{\text{audit}}$ be the audit group. If $\mathfrak{G}_{\text{audit}}$ is unsolvable (contains A_5), then there exist claims $x \in \mathcal{X}$ that cannot be audited by any algorithm. Symbolically:*

$$\mathfrak{G}_{\text{audit}} \text{ unsolvable} \implies \exists x \in \mathcal{X}, x \text{ unauditable.} \quad (1)$$

This is $P \Rightarrow Q$. The contrapositive is:

$$\forall x \in \mathcal{X}, x \text{ auditable} \implies \mathfrak{G}_{\text{audit}} \text{ solvable.} \quad (2)$$

Falsification strategy: Find a concrete claim x_0 and an audit algorithm that successfully audits it, while simultaneously proving that the audit group of the expert configuration contains A_5 . If such an x_0 exists, then by modus tollens on (??), SCX is refuted:

$$[\mathfrak{G}_{\text{audit}} \text{ unsolvable} \Rightarrow x_0 \text{ unauditable}] \wedge [x_0 \text{ auditable}] \implies \text{SCX is falsified.} \quad (3)$$

We call this the **Primary Falsifiable Prediction** (P_0): it requires both (a) constructing an expert configuration with A_5 audit group, and (b) demonstrating successful auditing on claims within that configuration. Either condition failing leaves SCX intact; both succeeding refutes it.

R 2. Crucial nuance: (??) is a **mathematical theorem**, not an empirical hypothesis. The falsifiable element is the claim that A_5 audit groups *exist in practice*. The theorem itself is deductively certain; the question is whether its antecedent can be realized.

1.3 Roadmap

Section 4 constructs the A_5 disagreement configuration. Section 7 presents the Honest Strike on A_5 realizability.

2 The Falsifiability Theorem

2.1 Formalizing the Verdict

We formalize audibility using the framework of Galois-SCX [?].

D 3 (A_5 -Containing Audit Group). Let $\mathcal{E} = \{1, \dots, n\}$ with audit group $\mathfrak{G}_{\text{audit}} \leq \text{Sym}_n$ (preserving the CEC). $\mathfrak{G}_{\text{audit}}$ **contains** A_5 if there exists a subgroup $H \leq \mathfrak{G}_{\text{audit}}$ with $H \cong A_5$ (the alternating group of order 60).

D 4 (Auditability — Operational Definition). A claim $x \in \mathcal{X}$ is **auditable**, denoted $\text{Auditable}(x) = \top$, if there exists an algorithm $\mathcal{A}$ and sample size N_0 such that for all $N \geq N_0$,

$$\P(\mathcal{A}(\text{data}_N(x)) = \text{truth}(x)) > \frac{1}{2} + \delta, \quad (4)$$

for some $\delta > 0$ independent of N .

D 5 (Falsifiability — Popperian). A theory T makes a **falsifiable prediction** P if there exists a possible observation O such that, were O to confirm $\neg P$ (i.e., $O \vdash \neg P$), T would be refuted.

T 6 (Galois Falsifiability — SCX is Falsifiable). Let T_{SCX} be the theory comprising Galois-

SCX. The following is a falsifiable prediction of T_{SCX} :

P_0 : **Primary Falsifiable Prediction.** There exists C_0 such that: $\mathfrak{G}_{\text{audit}}(C_0)$ contains A_5 .
Prediction: C_0 is unauditable: $\text{Auditable}(C_0) = \perp$.
Falsification condition: If any algorithm $\mathcal{A}$ returns $\text{Auditable}(C_0) = \top$ with probability exceeding $1/2 + \delta$ for some $\delta > 0$, then $(\text{Auditable}(C_0) = \top)$ contradicts the theorem, and is refuted.

Proof of Falsifiability. [**Rigorous Proof**] Let Γ be the axioms of T_{SCX} . By Galois-SCX, $\Gamma \vdash (A_5 \leq \mathfrak{G}_{\text{audit}}(C_0) \implies \neg \text{Auditable}(C_0))$. Let $E = (\text{Auditable}(C_0) = \top)$ be the empirical observation. Then:

$$\Gamma \vdash \neg \text{Auditable}(C_0), \quad E \vdash \text{Auditable}(C_0).$$

E contradicts Γ , so Γ is refuted (barring Duhem-Quine auxiliary hypotheses). Thus T_{SCX} makes a **falsifiable prediction**—the hallmark of a scientific theory. $\square$

R 7 (Popperian Integrity). • **Asymmetric falsifiability** (“risky prediction”): SCX makes a specific, concrete, *risky* prediction. Confirming it strengthens the theory; refuting it destroys it.

- **Auxiliary assumptions** (“modus tollens vs. Duhem-Quine”): The empirical realization of A_5 depends on auxiliary assumptions about expert behavior, measurement, and finite samples. Failure of the prediction could reflect failure of auxiliary assumptions rather than failure of the core theorem.

The P_0 prediction embodies the **Popperian spirit**—SCX “sticks its neck out” by specifying exactly what would prove it wrong.

2.2 Why P_0 Is a Genuine Test

P_0 satisfies all criteria for a genuine falsifiable prediction:

- (i) **Specific:** A_5 is precisely defined—“simple group of order 60,” “symmetries of the icosahedron,” **exactly 60 elements**. A_5 can be algorithmically verified (e.g., by checking that its subgroup lattice matches $\text{PSL}(2, 5)$). No “fuzzy” criteria.
- (ii) **Operational:** The prediction specifies an **exact performance barrier**—no audit algorithm can exceed $1/2 + \delta$ on A_5 -symmetric claims. This is a quantitative, measurable claim.
- (iii) **Risky:** The prediction is counterintuitive—most practitioners assume “more data” or “better algorithms” can solve any audit problem. SCX says they cannot, under specific, identifiable conditions.
- (iv) **Decisive:** A single counterexample—one algorithm achieving $> 1/2 + \delta$ accuracy on an A_5 -symmetric claim configuration—refutes the theory. This is “strong falsifiability.”

3 Three Falsifiable Predictions

Beyond the core P_0 , Galois-SCX makes two additional, independently testable predictions.

3.1 Prediction 1: The Quintic Invariant

Prediction 8 (Quintic Invariant Test). *Let $\mathcal{E} = \{E_1, \dots, E_5\}$ be five experts whose pairwise disagreement structure generates the full symmetric group $\mathfrak{S}_{\text{audit}} \cong S_5$ (hence containing A_5). Then:*

Any audit function $f : \{-1, +1\}^5 \rightarrow \{-1, +1\}$ on the expert verdicts must satisfy:

- (1) **Anonymity:** f must be symmetric under all permutations of experts;
- (2) **Neutrality:** f must treat $+1$ and -1 verdicts symmetrically;
- (3) **Monotonicity:** f must be monotone in the number of $+1$ verdicts;
- (4) **Quintic obstruction:** When exactly 2 experts return -1 and 3 return $+1$ (or vice versa), f cannot distinguish these configurations—it must output the same value.

Proof Sketch. [Rigorous Proof] Conditions (1)-(4) characterize S_5 -invariant functions on $\{-1, +1\}^5$. S_5 acts on $\{+1, -1\}^5$ by permuting coordinates. The orbits are classified by the number of $+1$ values, with orbit sizes $\binom{5}{k}$ for $k = 0, 1, \dots, 5$. There are exactly 6 orbits, corresponding to 6 possible S_5 -invariant functions $\{f_0, f_1, \dots, f_5\}$ where f_k outputs the “majority” based on k positive votes.

For $k = 2$ and $k = 3$, the orbits have size $\binom{5}{2} = \binom{5}{3} = 10$. An S_5 -invariant function cannot distinguish individual configurations within these orbits. In particular, the “2 vs. 3” ambiguity is the **quintic obstruction**: with 5 experts and a 3-2 split, no symmetric algorithm can determine which side is correct—the group action renders the two cases indistinguishable.

Any attempt to break this symmetry would require non- S_5 -invariant reasoning, which contradicts the premise that $\mathfrak{G}_{\text{audit}} = S_5$. $\square$

R 9 (Testability). Prediction ?? is directly testable: construct 5 experts with S_5 audit group, present them with claims yielding a 3-2 split, and verify that no S_5 -invariant audit algorithm beats random guessing.

3.2 Prediction 2: The A_5 Performance Obstruction

Prediction 10 (A_5 Performance Obstruction). *Let C be an expert configuration with $H \cong A_5 \leq \mathfrak{G}_{\text{audit}}(C)$. Then for any audit algorithm $\mathcal{A}$:*

“Even with infinite data”— $\mathcal{A}$ cannot achieve perfect accuracy:

$$\lim_{N \rightarrow \infty} \mathbb{P}(\mathcal{A}(\text{data}_N) = \text{truth} \mid C) = 1.$$

More precisely, there exists $\varepsilon_0 > 0$ (depending only on A_5) such that for any $\mathcal{A}$ and any N :

$$\mathbb{P}(\mathcal{A}(\text{data}_N) = \text{truth} \mid C) \leq 1 - \varepsilon_0. \quad (5)$$

The obstruction is absolute—it does not vanish with more data.

Proof Sketch. [**Rigorous Proof**] Since $H \cong A_5 \leq \mathfrak{G}_{\text{audit}}(C)$, the claim configuration C lives on an H -orbit $\mathcal{O}_H(C)$. A_5 acts on this orbit with degree at least 3 (since A_5 has no subgroups of index 2).

Any audit algorithm $\mathcal{A}$ must be H -invariant (symmetric with respect to relabeling experts): $\mathcal{A}(h \cdot \text{data}) = \mathcal{A}(\text{data})$ for all $h \in H$. The orbit $\mathcal{O}_H(C)$ has size at least 3, but the output space is $\{0, 1\}$ (binary verdict), so by the pigeonhole principle, at least two distinct truth values must map to the same output—**perfect accuracy is impossible**.

More formally, let χ be the character of the A_5 -representation on the data space. Decompose $\chi = \sum_i m_i \chi_i$ into irreducible characters. The trivial character χ_{triv} captures the invariant subspace. The maximum achievable accuracy is:

$$\mathbb{P}_{\text{max}} = \frac{1}{2} + \frac{1}{2} \cdot \frac{\chi_{\text{triv}}}{\chi(1)}.$$

Since A_5 has no nontrivial 1-dimensional representations, $\chi_{\text{triv}} < \chi(1)$ for any nontrivial representation, yielding $\mathbb{P}_{\text{max}} < 1$. The gap $\varepsilon_0 = \frac{1}{2}(1 - \chi_{\text{triv}}/\chi(1)) > 0$ establishes (??). $\square$

3.3 Prediction 3: Normalizer Violation

Prediction 11 (Normalizer Violation Test). *Let $\mathfrak{G}_{\text{audit}}(C)$ be the audit group of configuration C , and let $G_1, G_2 \leq \mathfrak{G}_{\text{audit}}(C)$ be two sub-teams. If*

$$G_1 \not\leq N_{\mathfrak{G}_{\text{audit}}(C)}(G_2) \quad \text{or} \quad G_2 \not\leq N_{\mathfrak{G}_{\text{audit}}(C)}(G_1), \quad (6)$$

then independent auditing is impossible. Specifically:

“Decomposed audits underperform joint audits”—

auditing G_1 and G_2 separately then combining produces systematically worse results:

$$\mathbb{P}_{\text{decomposed}}(\text{success}) \leq \mathbb{P}_{\text{joint}}(\text{success}) - \Delta,$$

where $\Delta > 0$ is the “coupling penalty” proportional to the normalizer violation.

Proof Sketch. **[Rigorous Proof]** By Galois-SCX [?], G_1, G_2 normalize each other $\iff$ they have direct product structure $\iff$ they are independently auditable. Violation of (??) means G_1 and G_2 do not normalize each other, hence their audit structures are coupled.

The coupling magnitude is quantified by the normalizer deficiency:

$$\delta(G_1, G_2) = \frac{|G_1 \setminus N_{G_1}(G_2)|}{|G_1|} \in [0, 1].$$

When $\delta > 0$, the decomposed audit incurs an information loss $\Delta = \Omega(\delta \cdot \gamma)$, where γ is the per-expert information content. $\square$

4 Experimental Design: Constructing an A_5 Audit Group

This is the operational core of SCX falsifiability.

4.1 Protocol for A_5 Disagreement Construction

Protocol 12 (A_5 Disagreement Construction). **Objective:** Construct a claim configuration C_0 such that $\mathfrak{G}_{\text{audit}}(C_0)$ contains A_5 .

Method:

Step 1: Expert pool: Select $n = 5$ experts (human or AI) with comparable expertise, denoted E_1, E_2, E_3, E_4, E_5 .

Step 2: Task design: Define a task $\mathcal{T} = \{T_1, \dots, T_K\}$ of binary classification problems. Each task T_k admits a “correct” answer—either positive (+1) or negative (−1).

Step 3: Calibration parameters:

- **Individual accuracy:** each expert has accuracy $p \in (0.5, 1)$;
- **Pairwise correlation:** $\text{Corr}(E_i, E_j \mid \theta) = \rho$ for all $i \neq j$;
- **No special structure:** e.g., E_1 is no more similar to E_2 than to E_3 .

Step 4: Symmetry: The joint distribution of $(E_1(x), \dots, E_5(x))$ is invariant under all permutations $\sigma \in S_5$. This ensures $\mathfrak{G}_{\text{audit}}(C_0) = S_5 \supset A_5$.

Step 5: Disagreement anchor: Introduce 5 “disagreement anchor” claims $\{x_1, x_2, x_3, x_4, x_5\}$ where expert E_i disagrees with the majority on claim x_i , inducing the 5-cycle disagreement pattern (1 2 3 4 5):

	x_1	x_2	x_3	x_4	x_5
E_1	−1	+1	+1	+1	+1
E_2	+1	−1	+1	+1	+1
E_3	+1	+1	−1	+1	+1
E_4	+1	+1	+1	−1	+1
E_5	+1	+1	+1	+1	−1

On each claim x_k , expert E_k alone dissents. The disagreement pattern is a 5-cycle C_5 , which generates S_5 (since C_5 together with any transposition generates all of S_5). Thus **the audit group is S_5 , which contains A_5 .**

Step 6: Task: The audit algorithm must determine, for each claim, whether the majority or the dissenter is correct.

4.2 Statistical Test

Falsification criterion: An algorithm $\mathcal{A}^*$ successfully audits $\{x_1, \dots, x_5\}$ if it achieves accuracy significantly above $1/2$ (e.g., at $p < 0.05$). If $\mathcal{A}^*$ achieves near-perfect accuracy (close to 100%), this would refute SCX—since SCX predicts that the A_5 obstruction makes this impossible.

4.3 Implementation Variants

- (A) **LLM-based.** Use 5 different LLMs or 5 differently prompted instances of the same LLM as experts.
- (B) **Synthetic.** Generate synthetic expert verdicts from a multivariate probit model with correlation matrix $\Sigma = (1 - \rho)I_5 + \rho\mathbf{1}_5\mathbf{1}_5^\top$. For $\rho > 0$, the audit group is S_5 since all permutations leave Σ invariant.
- (C) **Human experts.** Recruit 5 domain experts, calibrate their individual accuracies and pairwise correlations.

Statistical Decision Rule

Draw N bootstrap samples. Compute the empirical success rate $\hat{p}$.

H_0 (SCX holds): $p = 1/2$ (chance-level auditing).

H_1 (SCX is false): $p > 1/2$.

If $\sum_{k=\hat{p}N}^N \binom{N}{k} (1/2)^N < 0.05$, reject H_0 —SCX is falsified.

5 Complementarity with the Honest Person Theorem

5.1 Two Distinct Sources of Unauditability

Table 1: Galois vs. Honest Person Theorem

Dimension	Honest Person (Thm 3)	Galois Falsification
Core claim	Noise-difficulty indistinguishability $\implies$ unauditability	$\mathfrak{G}_{\text{audit}}$ unsolvability $\implies$ unauditability
Mechanism	Information-theoretic (Mutual Information, KL divergence)	Group-theoretic (orbit structure, representation theory)
Scope	Per-sample (applies to individual data points)	Structural (applies to entire claim configurations)
Obstruction	Semantics (what labels mean)	Symmetry (A_5 action on verdict space)
What it explains	That auditing fails (quantitative bound)	Why auditing fails (algebraic structure)

T 13 (Unification).

$$\text{Information-theoretic unauditability} \iff \mathfrak{G}_{\text{audit}} \text{ is unsolvable.} \quad (7)$$

*In words: **the information-theoretic indistinguishability proven by Theorem 3 is the Galois-theoretic manifestation of an unsolvable audit group**—the two perspectives are equivalent descriptions of the same underlying phenomenon.*

Proof Sketch. [**Rigorous Proof**]($\implies$) If noise and difficulty are information-theoretically indistinguishable from observational data, then the expert verdict patterns must admit symmetries that generate an unsolvable audit group—specifically, A_5 emerges as the automorphism group of the confusion structure.

($\impliedby$) If $\mathfrak{G}_{\text{audit}}$ is unsolvable, the orbit structure forces any audit function to be constant on orbits of size ≥ 3 , making noise-difficulty discrimination impossible. $\square$

R 14. This unification means P_0 tests **both** the Honest Person Theorem and Galois-SCX simultaneously—a single experiment with remarkable leverage.

5.2 “Information-Theoretic” vs. “Galois” Perspectives

Theorem 3 operates in the **information-theoretic** regime—it bounds what can be learned from finite samples. Galois-SCX operates in the **algebraic** regime—it characterizes the structural symmetries that make learning impossible. The A_5 obstruction is the bridge: when $\mathfrak{G}_{\text{audit}}$ contains A_5 , the information-theoretic bound becomes **absolute** (not merely finite-sample), because the group action renders verdict patterns fundamentally ambiguous.

6 Practical Guidance: Galois Pre-Audit for SCX Deployment

Before deploying SCX, practitioners should verify that the expert configuration does not contain an A_5 obstruction—a **pre-audit audit**.

6.1 Protocol: Galois Pre-Audit

Protocol 15 (Galois Pre-Audit). **Input:** Configuration C with expert set $\mathcal{E}$ and claim space.

Output: Risk assessment: Safe / Warning / Critical.

Procedure:

D1 Compute pairwise disagreement. For all pairs (i, j) , estimate $d_{ij} = \mathbb{P}(E_i(x) \neq E_j(x))$.

D2 Construct disagreement graph. Build a weighted complete graph on experts with edge weights d_{ij} . Compute its **automorphism group** $\text{Aut}(\Gamma)$ —permutations that preserve pairwise disagreement structure. $\text{Aut}(\Gamma)$ is the empirical estimate of $\mathfrak{G}_{\text{audit}}$.

D3 Detect A_5 . Check whether $\mathfrak{G}_{\text{audit}}$ contains a subgroup isomorphic to A_5 . For $n \leq 10$, this is computationally feasible via exhaustive subgroup enumeration; for larger n , use the fact that $\mathfrak{G}_{\text{audit}} \geq A_5$ iff $\mathfrak{G}_{\text{audit}}$ is non-solvable (which can be tested via composition series).

D4 Risk classification.

- $A_5 \leq \mathfrak{G}_{\text{audit}}$: **Critical**— A_5 obstruction present; SCX auditing cannot guarantee correctness.
- $\mathfrak{G}_{\text{audit}}$ non-solvable but no A_5 detected: **Warning**—non-solvable audit group; reliability not guaranteed.
- $\mathfrak{G}_{\text{audit}}$ solvable: **Safe**—audit is theoretically guaranteed.
- Indeterminate (insufficient data): **Inconclusive**—collect more calibration data.

R 16 (Computational Feasibility). For $n \leq 10$, automorphism group computation is feasible ($10! = 3,628,800$, tractable with modern algorithms). Beyond $n = 10$, heuristic methods based on spectral properties of the disagreement matrix can detect A_5 without full group enumeration.

6.2 Remediation Strategies

If the Galois pre-audit detects an A_5 obstruction:

- (1) **Add experts.** The symmetric group Sym_n (for $n \geq 5$) always contains A_5 . Adding experts without changing the correlation structure does not resolve the obstruction.
- (2) **Break symmetries.** Introduce asymmetric expert weights, domain-specific calibration, or partial ordering. The key is to ensure pairwise disagreement patterns are not all identical—this breaks the S_5 symmetry.
- (3) **Stratify the claim space.** Restrict auditing to subdomains where the expert disagreement graph has a solvable automorphism group.

7 Honest Strike — Can A_5 Be Realized in Practice?

This is the central, honest challenge to SCX. The entire falsifiability framework hinges on whether A_5 audit groups can actually be constructed.

7.1 Strike 1: The Difficulty of Perfect Symmetry

[Honest Strike]Real expert panels are never perfectly symmetric.

The protocol in Section 4 requires $\mathfrak{G}_{\text{audit}} = S_5$, a group of order 120. Achieving this requires:

- **Equal individual accuracies** across all 5 experts;
- **Equal pairwise correlations** $\binom{5}{2} = 10$ constraints;
- **No higher-order structure** (three-way, four-way interactions).

LLM implementation: Prompt engineering can approximately achieve this, but exact equality is impossible with finite samples. The audit group will be $\mathfrak{G}_{\text{audit}} = \{\text{id}\}$ rather than S_5 , because real data always contains asymmetries.

Response: The prediction does not require *exact* symmetry. Approximate S_5 -symmetry suffices: if the departure from symmetry is ϵ -small, the A_5 obstruction is weakened but not eliminated. The key question becomes quantitative: how much asymmetry is needed to break the obstruction?

Strike resolution:

S_5 is a mathematical idealization. Real expert panels approximate it. The A_5 obstruction is **robust under small perturbations**—it does not require perfect symmetry, only sufficient symmetry to embed A_5 . SCX’s prediction is **quantitatively falsifiable**: specify the symmetry tolerance ϵ .

7.2 Strike 2: The “ A_5 ” May Not Be What We Think

[Honest Strike]Does “ A_5 in the audit group” mean what we think it means? What if a group isomorphic to A_5 appears for “trivial” reasons?

If $\mathfrak{G}_{\text{audit}} = S_5$, then of course $A_5 \leq \mathfrak{G}_{\text{audit}}$ (since $A_5 \trianglelefteq S_5$). But this is a **logical consequence of the definition**, not an empirical discovery. The real question is whether non- S_5 audit groups containing A_5 can arise from expert behavior.

[Open Problem]Stability of A_5 under noise. If $\mathfrak{G}_{\text{audit}}$ is approximately S_5 (within ϵ), does it still contain a subgroup approximately isomorphic to A_5 ? What is the “approximate subgroup” theory for audit groups? This connects to **stability theory in group theory**—when does an approximate homomorphism imply a nearby exact homomorphism?

7.3 Strike 3: The 5-Cycle Disagreement Has Only D_5 Symmetry

[Honest Strike]The 5-cycle disagreement pattern in Protocol ?? (where E_k dissents on x_k) has automorphism group D_5 (order 10), not S_5 (order 120), and not A_5 (order 60).

The pattern where each expert dissents on exactly one claim:

- $(1\ 2\ 3\ 4\ 5)$ is a symmetry (rotating which expert dissents on which claim);
- $(1\ 5\ 4\ 3\ 2)$ is also a symmetry (the reverse cycle);
- Reflection about any expert (flipping the cycle) is a symmetry.

These generate D_5 (order 10), which is **solvable**—it has normal subgroup $\mathbb{Z}/5\mathbb{Z}$ with quotient $\mathbb{Z}/2\mathbb{Z}$. So this particular pattern does **not** realize A_5 !

Correction: The 5-cycle pattern alone does not suffice. To realize A_5 , we need the full symmetric group S_5 , which requires the disagreement pattern to be invariant under *all* permutations, not just cyclic ones. This means:

- **All 5 experts must have identical individual accuracies;**
- **All 10 pairwise correlations must be identical;**
- This is achieved by the “exchangeable expert” design in Section 4.3(B), where the correlation matrix is $\Sigma = (1 - \rho)I_5 + \rho\mathbf{1}_5\mathbf{1}_5^\top$.

The 5-cycle pattern is then **embedded within** the full S_5 configuration as a specific disagreement realization—but the audit group itself is S_5 , not D_5 .

7.4 Strike 4: The Algorithm Invariance Assumption

[Honest Strike]Galois-SCX assumes the audit algorithm is invariant under the audit group. But real algorithms need not respect this invariance.

The proof assumes $\mathcal{A}(\sigma \cdot \text{data}) = \mathcal{A}(\text{data})$ for all $\sigma \in \mathfrak{G}_{\text{audit}}$. **This is a strong assumption.** Real algorithms can break the symmetry by, e.g., “trust E_3 more than E_1 ” based on historical performance. If the algorithm breaks the symmetry, the A_5 obstruction may dissolve.

Response: The invariance is not an assumption—it is a **consequence** of not knowing which expert is which. If the algorithm has side information that distinguishes experts (e.g., past accuracy), then the effective audit group is smaller and may be solvable. But this side information must itself be **auditable**—how do we know expert E_3 is more reliable? That meta-audit faces its own A_5 problem.

Strike: In practice, calibration data always provides side information that breaks symmetry. The A_5 obstruction is a **worst-case bound** that may not bind in typical scenarios.

8 Conclusion

8.1 Summary

- (1) **SCX is falsifiable.** We have identified a concrete, experimentally testable prediction (P_0) that would refute SCX if confirmed. The A_5 obstruction is the fulcrum.
- (2) **Three falsifiable predictions:** (a) the primary modus tollens (P_0); (b) the A_5 performance obstruction (accuracy < 1); (c) the normalizer violation test.
- (3) **Experimental protocol exists.** We have designed an explicit $n = 5$ expert disagreement configuration and statistical test that can, in principle, falsify SCX.

- (4) **Honest limitations.** The realizability of exact A_5 audit groups in practice is the central open question. SCX’s falsifiability is genuine but conditional on approximate A_5 realizability.

8.2 The Path Forward

SCX invites falsification.

- (1) SCX makes a **risky prediction**—that A_5 -symmetric expert configurations are unauditable.
- (2) SCX specifies the **exact conditions** under which it would be wrong—successful auditing of an A_5 configuration.
- (3) SCX provides the **experimental blueprint** (Section 4) and the **statistical decision rule** for the test.
- (4) If the experiment succeeds in falsifying SCX, A_5 would become the **most important counterexample** in audit theory—and SCX would be refuted with honor.

8.3 Open Problems

The Honest Strike—the A_5 realizability question—is the most urgent open problem: **[Open Problem]Stability of A_5 under perturbations.** How much asymmetry (in accuracy, correlation) can an S_5 -intended configuration tolerate before the A_5 obstruction dissolves?

[Open Problem]Constructive A_5 realization. Provide an explicit, reproducible protocol that demonstrably yields an audit group containing A_5 , with rigorous “closeness” guarantees.

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